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Oil and Gas Field Digital Twin Full Lifecycle Management Platform and Application Based on Meta-Universe Technology

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Abstract

Oil and gas fields are often characterized by wide distribution of facilities and equipment, harsh natural environments, difficulties in engineering construction and production operation and maintenance management, and increasingly severe risks of violence and terrorism. There is an urgent need to build a unified management platform covering the engineering construction, production and operation phases, and to carry out unified control over the whole life cycle process of oilfields. This paper takes meta-universe technology as the core to construct a virtual reality environment for oil and gas field production and office facilities. It integrates the static data in the engineering construction process, such as engineering design, procurement, construction, commissioning, and startup, with the dynamic data and business processes in the production and operation process, such as Internet of Things (IOT) production, sales and procurement, and comprehensive office, to build a digital twin full life cycle management platform for oil and gas fields, and applies it to an oil field in West Africa. Aiming at the problem of poor network transmission conditions, this paper researches the adaptive dimensionality reduction algorithm for digital twin high-dimensional animation data, improved Forward Error Correction coding (FEC) and Automatic Retransmission Request (ARQ) algorithms, and adaptive transmission algorithms based on active sampling of the "protagonist aura" model. This ensures stable operation of the twin platform and its applications in restricted and highly unstable network environments. In terms of equipment dynamic fault identification and predictive maintenance, this paper adopts high-fidelity modeling and multi-case simulation technology to realize the modeling and simulation of reciprocating compressors, and establishes a fault diagnosis model based on machine learning and deep learning, which is capable of accurately identifying and classifying different types of faults and providing early warning, with a fault identification accuracy rate of up to 95% and an early warning accuracy rate of up to 90%. In general, the working mode of the oilfield is transformed from "labor-intensive, stationed on duty, regular inspection" to "fewer or unmanned, centralized monitoring, on-demand inspection". The management mode changes from"manual judgment + experience analysis" to "automatic warning + intelligent analysis". The abnormality handling mode is transformed from "real-time alarm and post-disposal" to "pre-warning and mid-control". Through the construction of the platform, it improves

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the company's data governance capability, optimizes the operation and management process, improves the production operation efficiency, directly reduces the labor cost and operation cost by about 37%, improves the efficiency of production operation fault disposal to 75%, reduces the number of production emergencies by more than 30%, and realizes the full life-cycle management of the oilfield.

Keywords: Meta-Universe, Digital Twin, Smart Oil and Gas Field, Total Lifecycle Management

Introduction

With the increasing global energy demand, efficient management of oil and gas fields has become an important issue in the oil and gas industry. The traditional management methods rely on experience and historical data, which make it difficult to cope with the complex and changing underground environment and equipment operating conditions, as well as the realistic and complex production environment. For example, the production base of a company in PetroChina West Africa has a harsh natural environment, low social dependence, and high risk, and the production operation is in the mode of "fixed watch and regular inspection", the operation management is in the mode of non-interoperability of data from multiple systems, and the operation decision-making is mostly based on the decision-maker's experience.

The emergence of meta-universe technology and digital twin technology provides a new opportunity for the whole life cycle management of oil and gas fields. By building a virtual oil and gas field environment, managers can monitor the operating conditions of the oil and gas field in real-time, make simulation predictions, optimize the production process, warn of equipment failures in advance, and improve production efficiency and safety. In addition, meta-universe technology can also provide immersive training environments to help employees quickly master operational skills and improve work efficiency.

This paper aims to discuss the design and realization of the digital twin full life cycle management platform for oil and gas fields based on meta-universe technology. By integrating Internet of Things (IoT), big data, artificial intelligence and virtual reality technologies, a virtual oil and gas field environment is constructed to realize the full lifecycle management of oil and gas fields from exploration, development to production and maintenance. The innovation of this research is to provide a highly immersive management and decision-support environment using meta-universe technology to improve the intelligence of oil and gas field management.

Current Status of Research

Although meta-universe technology and digital twin technology show great potential in oil and gas field management, their application still faces many challenges. First, real-time data collection and transmission is a key issue that requires efficient IoT technology support. Second, the storage and processing of big data require powerful computing power and advanced analyzing technology. In addition, constructing high-precision digital twin models requires a large amount of data and expertise, and the accuracy and real-time performance of the models directly affect the effectiveness of decision-making.

Industrial IoT technology can realize the comprehensive monitoring of the production process of oil and gas fields through the sensor network, which provides the basis for data collection and analysis. Big data technology can dig out the laws and knowledge hidden behind the data through the processing and analysis of massive data, providing decision support for managers [1]. In oil and gas field management, the combined application of IoT and big data technology not only improves the efficiency and accuracy of data processing, but also promotes the intelligence and refinement of management [2].

Artificial intelligence (AI) technology shows great potential in the field of fault diagnosis with its powerful data processing and pattern recognition capabilities. By training fault diagnosis models, automatic identification and early warning of equipment faults can be realized. For example, in the oil and gas field, feature extraction and classification of vibration signals using deep learning algorithms can accurately

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identify the fault types of reciprocating compressors [3]. This data-driven fault diagnosis method not only improves the accuracy and efficiency of fault identification, but also mitigates reliance on manual expertise and experience [4].

In summary, in order to cope with the current management obstacles, it is particularly urgent and important to design and develop a digital twin full lifecycle management platform for oil and gas fields based on meta-universe technology by combining industrial IOT, big data, AI and virtual reality (VR).

Oil and Gas Field Digital Twin Full Lifecycle Management Platform Based on Meta-Universe Technology

With digital twin as the core, this paper builds a virtual reality environment for production and office facilities, integrating static data during engineering design, procurement, construction, trial operation and commissioning, and dynamic data during production, sales and procurement, and comprehensive office operation. It forms the capabilities of business application integration and collaboration, data management from construction to delivery, visual operation control, project implementation process management, procurement business management and decision-making, etc., and forms a smart oil and gas field with "comprehensive perception, automatic control, trend prediction, intelligent optimization and collaborative operation".

Technical Route

The platform is based on a hyper-converged computer network architecture, with unified planning and construction of the infrastructure for hosting software platforms and applications. Information system integration technology is applied to realize the integration of the twin platform with various business application systems through application interfaces. On the digital twin platform, rapid modeling technology is applied to realize the twin construction of digital employees, office environments and major equipment, the twin creation of business activities based on workflow engine, video conferencing and instant messaging applications, and the management of models, documents, attributes and their associations of factory objects such as production facilities and equipment based on digital delivery applications. On the basis of the above digital mapping of production and operation, real-time data of production IoT is collected through Object Linking and Embedding (OLE) for Process Control (OPC) interface, dynamic data of sales and purchasing and operation management application is collected through human-computer interaction, and data mapping between twin space and physical space is realized by using rule engine, workflow engine and 3D visualization engine.

Troubleshooting and health status monitoring services for reciprocating compressors and key machine pumps are deployed on the hyper-converged platform through virtualization technology respectively. The production IoT Supervisory Control And Data Acquisition (SCADA) data is collected in real time through the OPC interface, the machine learning and deep learning equipment health status assessment and fault diagnosis models are applied to identify the key equipment status, and the digital twin platform application is docked through the Application Programming Interface (API) to realize the visualization display of the key equipment status.

The platform seamlessly integrates application bandwidth adaptive scheduling with client-side caching and fusion technology, improved Forward Error Correction coding (FEC) and Automatic Retransmission Request (ARQ) algorithms, Principal Component Analysis (PCA)-based dimensionality reduction algorithms, "protagonist aura" active sampling, and layered adaptive coding technology, transmission protocols, and more. It is applied to business office digital twin application scenarios, which can realize smooth interaction of complex 3D models in network bandwidth-constrained environments and enhance the interaction experience.

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General Architecture

The overall architecture of the platform is shown in Figure 1 and consists of one center (Display Collaboration Center) and three middle platforms (Data, Technology and Business Middle Platforms).

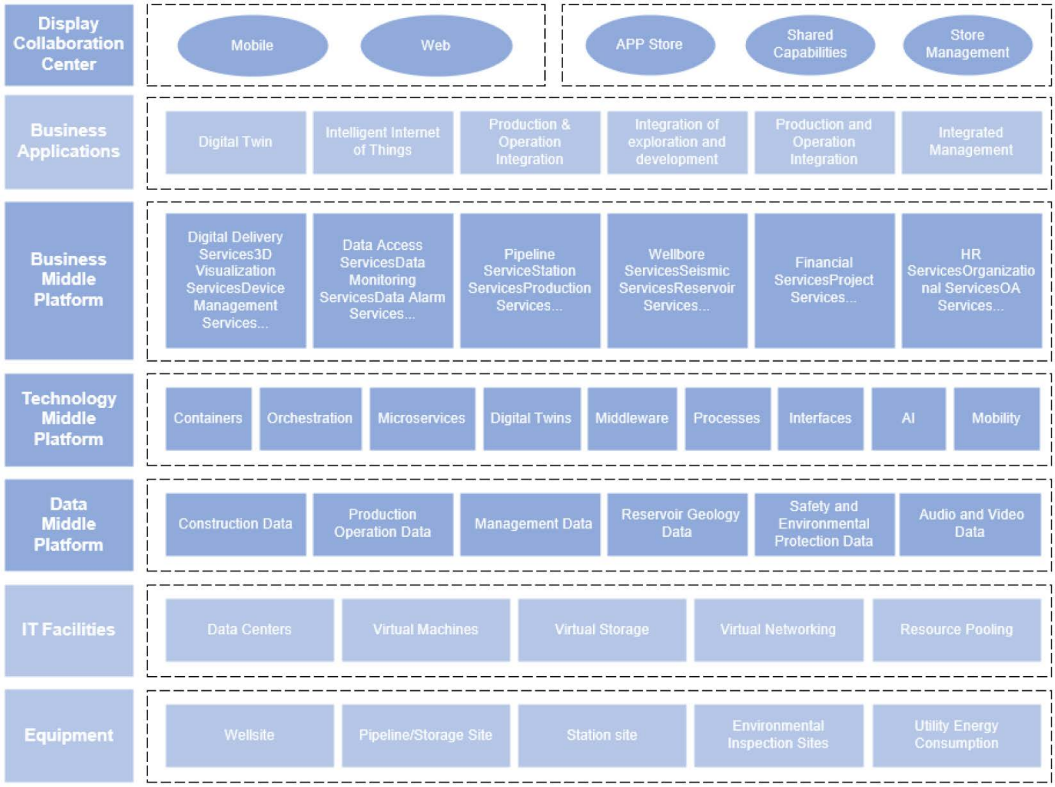

Figure 1—Overall Architecture

The Data Middle Platform is responsible for aggregating and integrating a myriad of data sources encompassing engineering construction, production operations, operational management, reservoir geology, safety and environmental protection, as well as audio-visual information. It facilitates data sharing, fusion, organization, processing, modeling, analysis, governance, and service applications, thereby establishing a comprehensive data capability platform that standardizes data formats and offers services through API interfaces.

The Technology Middle Platform integrates advanced and proven technologies such as digital twins, intelligent IoT, and artificial intelligence, leveraging the robust Docker+Kubernetes PaaS infrastructure. This platform provides the foundational technical resources and capability support necessary for the Business Middle Platform and applications.

The Business Middle Platform, grounded in the Data Middle Platform, meticulously abstracts and reconceptualizes the critical functionalities of oilfield business applications. By encapsulating these functionalities into reusable capabilities, it fosters a robust and versatile platform that promotes efficiency, scalability, and innovation in the domain of oilfield operations

Core Technology Research

Equipment Intelligent Operation and Maintenance

In order to realize intelligent equipment operation and maintenance, high-fidelity modeling and multi-conditional simulation techniques are adopted to realize the modeling and simulation of reciprocating compressors. Moreover, a fault diagnosis model based on machine learning and deep learning methods is

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established, which can accurately recognize and classify various fault types and issue early warnings, and the fault recognition accuracy of the model is as high as 95% and the warning accuracy is as high as 90%.

First, according to the research framework shown in Figure 2, this paper establishes a digital intelligent monitoring system using shape-attribute mapping visualization technology. Aiming at the problem that the mechanism model for combining compressor monitoring and warning technology is still immature, the system uses high-fidelity modeling and multi-case simulation technology to model a real reciprocating compressor. Using control logic mapping technology, the system continuously collects real-time data from the equipment, processes and analyzes it at high and low frequencies, and extracts key feature information, including sensor data such as temperature, pressure, and flow rate, as well as operating parameters and performance indicators of the equipment, with a fidelity of 90%. In addition, Proportion Integration Differentiation (PID) process mapping technology is used to present the processed data in the form of dynamic charts, graphical process diagrams, and real-time animated simulations, thereby increasing the response speed to abnormal conditions by approximately 30%.

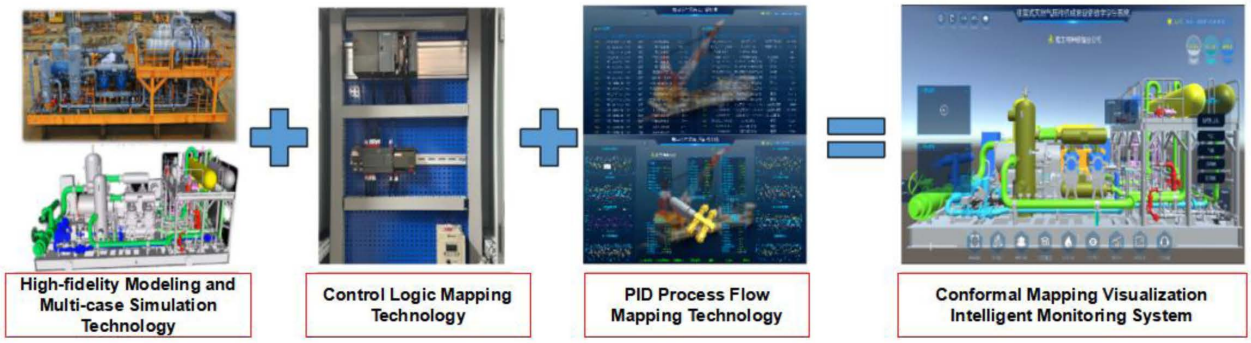

Figure 2—Development of Digital Intelligent Monitoring System with Shape-Attribute Mapping Visualization Technology

Second, a method for calculating data from multiple virtual sensors based on data analysis was developed (shown in Figure 3). The method customizes high-frequency and low-frequency data acquisition and transmission schemes based on the sampling frequency required by each sensor, aiming to reduce data redundancy and improve the real-time and stability of data transmission. A comprehensive virtual sensor test methodology was established to validate the stability and real-time data transmission capability of the system. The validation process involves simulating compressor operation and monitoring data through actual sensors, thus validating the real-time performance of the data acquisition and transmission scheme and virtual sensors, and ultimately reducing the number of sensor deployment points.

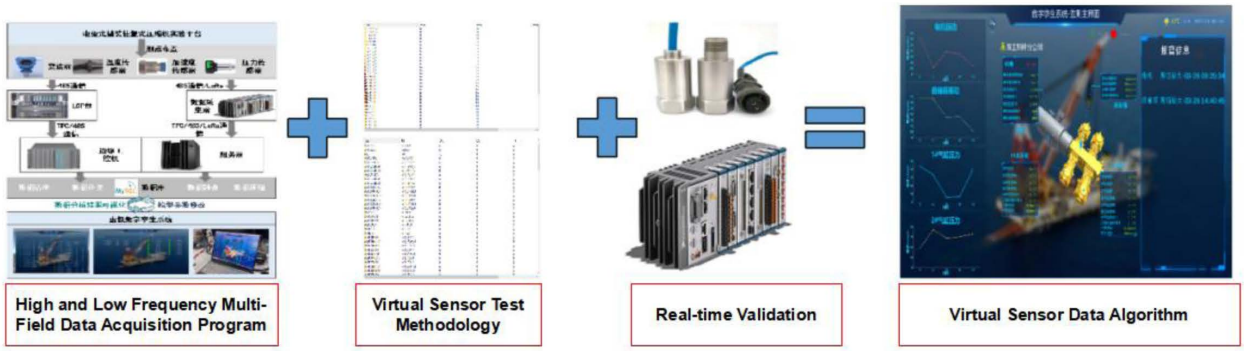

Figure 3—Development of Multi-virtual Sensor Data Arithmetic Method

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Finally, a fault identification and warning algorithm based on "health state + operation condition data" is established (as shown in Figure 4). A Support Vector Data Description (SVDD) algorithm is used for health state identification to construct a dataset to monitor and evaluate the health state of the equipment. The received data are organized into an operational condition database and then preprocessed to extract fault-related feature information including statistics, spectral features, and time-domain features. Representative and discriminative features are selected using machine learning algorithms, on the basis of which machine learning and deep learning based fault diagnosis models are constructed. These models are trained with annotated historical data so that they can learn the features of various fault modes from the data and accurately identify, classify and warn about different types of faults.

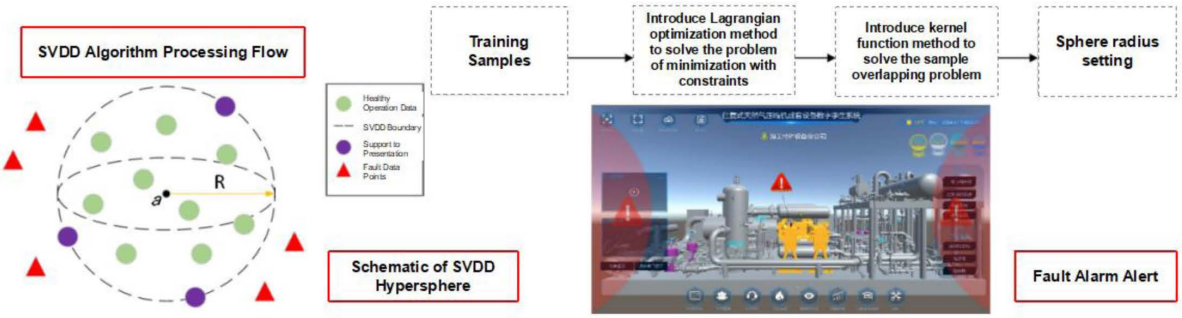

Figure 4—Development of Fault Identification and Early Warning Algorithm

Adaptive Dimension Reduction Data Transmission Technique Based on "Protagonist Halo"

Aiming at the challenges of data transmission in low-bandwidth and highly fluctuating network environments, this paper proposes a solution that integrates coding theory, data compression, and adaptive transmission techniques to improve data transmission performance.

First, customized ARQ and FEC strategies are designed to meet the higher requirements of digital twin technology for real-time, consistency and synchronization of data:

- Transmitter forward error correction coding: Improve the error correction capability during channel data transmission by introducing moderate coding redundancy.
- Receiver error correction decoding: Uses low-complexity, high-efficiency decoding algorithms to quickly detect and correct transmission errors.
- Adaptive retransmission strategy: Dynamically adjust the retransmission trigger threshold according to channel conditions to optimize transmission reliability and efficiency.
- Data prioritization: Different retransmission strategies and coding redundancy are applied to critical and non-critical data.

Traditional ARQ and FEC methods usually rely on preset thresholds and rules for retransmission and error correction, which are difficult to dynamically adapt to the complex and changing network environment. In this paper, we introduce a machine learning algorithm that autonomously adjusts parameters such as coding redundancy and retransmission triggering thresholds by learning the network state and data transmission quality in real time to achieve intelligent adaptive optimization of transmission strategies. This machine learning-based approach can better adapt to the dynamic network changes in digital twin applications and provide more stable and reliable data transmission services.

In digital twin applications, the transmission quality requirements of different types of data vary greatly. In this paper, a multi-level quality assurance mechanism is developed to adopt different transmission strategies for different levels of data. For critical data, higher coding redundancy and stricter retransmission strategies are used to ensure transmission reliability; for non-critical data, lower redundancy and looser

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retransmission strategies are used to reduce transmission overhead. This multi-level quality assurance improves the overall transmission efficiency and guarantees the transmission quality of critical data.

Second, to address the challenge of managing data transmission volume, especially when dealing with high-dimensional information such as skeletal animation data in digital twin applications, this paper develops a PCA-based dimensionality reduction transmission technique. Through accurate data processing, this technique greatly improves the data transmission efficiency and reduces the network burden at the same time:

- High-dimensional data characterization: PCA dimensionality reduction technique is applied to the high-dimensional characteristics of digital twin skeletal animation data.
- Adaptive dimensionality reduction strategy: Dynamically adjust the target dimensions of dimensionality reduction according to network bandwidth and data importance.
- End-to-end system design: Seamlessly integrate PCA dimensionality reduction algorithms with data encoding and transmission protocols.

PCA as a statistical technique extracts the most important features from multidimensional data and reduces the data dimensionality without losing key information. In digital twin skeletal animation, this effectively compresses the raw data volume and minimizes the amount of data processing during transmission. This speeds up the data transmission and alleviates the problem of delay and bandwidth consumption due to the large amount of data.

The seamless integration of PCA dimensionality reduction algorithms with data encoding and transmission protocols enables end-to-end system design from data acquisition to processing and transmission. This integration simplifies the application process and improves system collaboration and efficiency. The end-to-end design ensures that every step from the data source to the final reception is optimized, guaranteeing the efficient operation of every part of the data processing and transmission process. This not only improves the efficiency of processing and transmitting digital twin skeletal animation data, but also ensures accurate and fast transmission of critical data in complex and changing network environments.

Finally, the "protagonist highlight" mode realizes high-precision and high-fidelity sampling of the core content of users' concerns through active sampling for adaptive transmission:

- Protagonist highlight sampling: High-precision and high-fidelity sampling of the core content of user concerns.
- Peripheral scene simplification: Low sampling accuracy and fidelity for non-critical areas.
- Hierarchical adaptive encoding: Encoding and transmission of data at different levels according to priority.
- Adaptive bandwidth scheduling: Dynamically adjust the data encoding bit rate and transmission strategy according to network bandwidth conditions.

By applying "protagonist halo" sampling technology, the system is able to identify and prioritize the most important data to ensure its high-quality transmission. For example, in a digital twin conference, key visual information such as the main visuals of the speaker's voice and actions are transmitted with the highest fidelity, ensuring that the receiver receives the most accurate information. For non-core or less user-focused content, lower sampling accuracy and fidelity are used to significantly reduce the amount of data for non-critical content, thus reducing the overall network burden and transmission cost.

The hierarchical adaptive coding strategy further optimizes data flow management by ensuring that the most important information is always transmitted first under any given network conditions, while adjusting the transmission strategy for other data according to real-time changes in network bandwidth. This dynamic adjustment is particularly suitable for unstable network environments, such as mobile communications or remote regional connections, effectively ensuring the continuity and efficiency of data transmission.

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Application results

The research results have been successfully applied to an oil field operated by PetroChina in West Africa. A dual platform for oil and gas field production and operation based on meta-universe technology was established during the construction of oil field surface facilities. The implementation of the platform and its application has promoted the transformation of the working mode of the oilfield from "labor-intensive, on-site manpower dispatching and regular inspection" to "less or no manpower, centralized monitoring and on-demand inspection". In addition, it transforms the management method from "manual judgment + empirical analysis" to "automatic warning + intelligent analysis", and changes the abnormality handling mode from "real-time alarm and after-action response" to "pre-warning and process control". These changes have enhanced the company's data governance capability, optimized the operation and management process, improved production and operation efficiency, and realized the full life cycle management of the core assets of the oilfield.

Engineering Construction - Digital Delivery and Engineering Project Management

Data generated during oilfield construction, including models, PIDs, documents, attributes, etc., are digitally delivered to form a unified and efficient data asset, which lays a high-quality data foundation for the full realization of a smart oilfield. In addition, the platform can incorporate construction, progress, quality, procurement, Quality, Health, Safety and Environment (QHSE) and other related data into the database for management. As shown in Figure 5, the platform uses digital twin technology to build a visual interface that can intuitively display construction progress, control construction progress, adjust dynamic resources and rationalize project schedules.

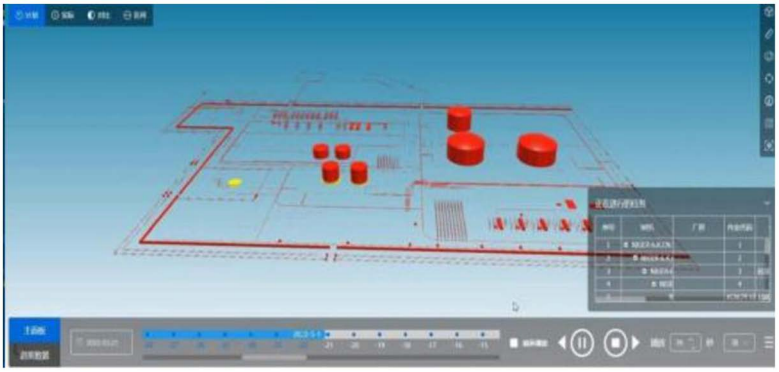

Figure 5—Engineering Project Management Interface

Production Operation - Unmanned / Less-manned Operation

The platform is based on the IoT system and uses optical transmission and oilfield Local Area Network (LAN) as communication channels to realize unmanned management of well sites, multiphase pumping stations, and metering stations, as well as less-manned operation of joint stations, dehydration stations, and transfer stations.

Production Operations - Visualizing Single Well, Station and 3D Pipeline Management

The Visual Single Well Management solution integrates SCADA and Virtual Flow Metering (VFM) system data with a digital twin foundation to automatically collect data such as operating currents, voltages, and vibrations from wellsite equipment, enabling real-time monitoring of operating conditions. It also displays real-time production data, including fluid levels, pressures and temperatures, to aid in production decisions. Historical data is automatically organized into integrated well history data for easy prediction of production trends.

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Visualizing site operations and maintenance relies on simulated three-dimensional (3D) site models, combined with real-time data from SCADA and Closed Circuit Television (CCTV) systems, to display real-time (and historical) field production data, provide fire and gas alarm information, monitor the status of automated valves and rotating equipment, and guide on-site production and maintenance operations. As shown in Figure 6, tracking station operation information and safety status helps safety management.

(a) Vessel Levels (b) Production Monitoring

Figure 6—Interface for Visualizing Station Management

In addition, with the 3D Geographic Information System (GIS) engine, facilities such as wells, stations, off-station pipelines, transmission lines, and communication cables can be displayed in the 3D GIS, providing comprehensive support for the visualization and management of oilfield surface production.

Production Operations - Visualizing Equipment Operation and Maintenance

As shown in Figure 7, high-fidelity 3D equipment models integrated with SCADA and CCTV systems provide real-time display of key equipment operating parameters and assessment of their operational status. Real-time access to information around the equipment creates a model and data foundation for intelligent equipment management. By combining the equipment condition monitoring model, the platform can diagnose the health status of key equipment, thus improving the overall level of intelligence.

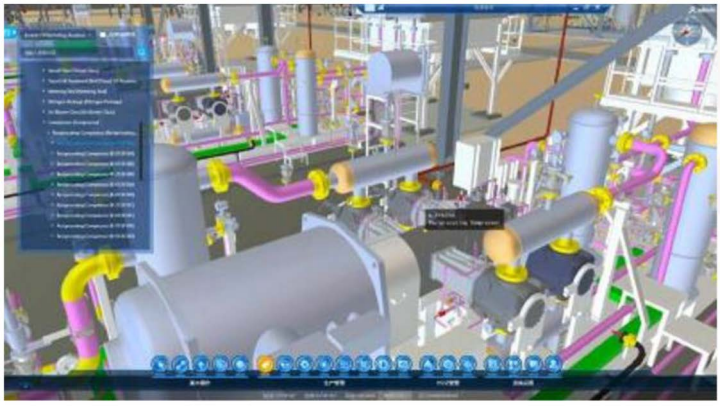

Figure 7—Interface for Visualizing Equipment Operation and Maintenance

Business Decision Making - Collaborative Office, Intelligent Procurement and Decision Support Center

The platform is centered on workflow tasks and promotes collaborative working in a virtual environment through task pushing, forecasting, processing and archiving. As shown in Figure 8, digital twins are utilized to create virtual roles and meetings for "face-to-face" communication in virtual environments, enabling immersive communication and enhanced user experience.

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Figure 8—Collaborative Office Interface

It establishes a human-centered, procurement process-driven intelligent procurement management platform scenario, realizes refined management of procurement business, effectively improves production and office efficiency, reduces manpower and system development costs, transforms the management mode, and improves the quality of management.

The decision support system is management-oriented, covering the whole business chain from business activities such as planning and finance to production activities such as oil extraction and pipeline transportation. It provides business departments with a unified intelligent management cockpit and dashboard, which improves the speed of information transmission, reduces costs and improves overall decision-making efficiency.

Conclusion

Integrating meta-universe and digital twin technologies into oil and gas field management provides a comprehensive, intelligent and immersive approach to managing complex operations. The development and application of the oil and gas field digital twin full lifecycle management platform addresses the limitations of traditional management methods and adapts to the dynamic and challenging conditions of modern oil and gas fields.

The platform architecture consists of a display collaboration center and three middle platforms (data, technology, and business) that effectively leverage IoT, big data, AI, and VR technologies to support integrated data management, technology resources, and business functions. The platform is able to simulate high-fidelity models and provide accurate fault diagnosis, fully demonstrating its advanced capabilities in intelligent operation and maintenance. In addition, the implementation of adaptive degradation data transmission technology ensures reliable data communication even in low-bandwidth and highly fluctuating network environments, further enhancing the robustness of the platform.

The successful application and initial deployment of the platform in West African oilfields validates its efficacy and provides a compelling case for subsequent rollout.

Acknowledgments

Sincere thanks are due to all the leaders and experts who provided guidance and assistance during the research process of the platform. Their invaluable support enabled the initiation and smooth progress of this innovative project. Meanwhile, sincere thanks to the research team, whose unremitting efforts and intellectual contributions have played a key role in the success of this project. The team members worked together, faced the challenges head-on, and overcame the technical difficulties to successfully apply metadata technology to oil and gas field management, realizing the leap from theory to practice.

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In addition, special thanks go to the cooperative application units of the project. Their trust and cooperation have promoted the application of the research results in the oil and gas industry and provided strong support for improving production efficiency and management level.

Finally, it is sincerely hoped that the platform will continue to flourish and make significant contributions to the digital transformation and high-quality development of the oil and gas industry.

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